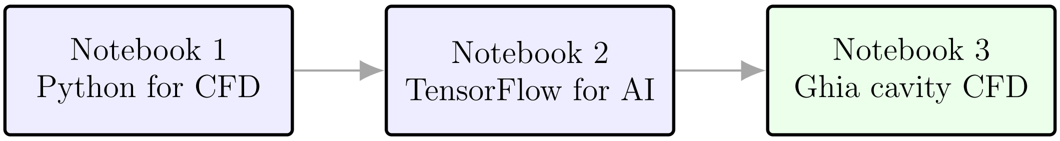
AI in Fluids – Week 1

Introductory Fluid Mechanics for Students New to CFD

A simplified lecture note aligned with the Python, TensorFlow, and Ghia cavity notebooks



Course idea: first learn basic coding, then learn ML concepts, then apply them to one clean CFD benchmark.

Course theme: learn the physics, learn the code, then learn from data.

This Week-1 version uses finite differences and the streamfunction–vorticity formulation.

Lattice Boltzmann and advanced CFD solvers are intentionally not part of this Week-1 package.

Contents

1 Purpose of this lecture

This course is called **AI in Fluids**. The word "AI" does not remove the need to understand fluid mechanics. A neural network can fit data, but it cannot tell us by itself whether a flow field conserves mass, satisfies a wall boundary condition, or matches a trusted benchmark. Therefore, the first week has a simple goal: build a shared language for fluids, Python programming, tensors, and one basic CFD benchmark.

The first week is organized around three notebooks:

1. **Python for CFD**: variables, loops, functions, NumPy arrays, finite differences, residuals.
2. **TensorFlow for AI in Fluids**: tensors, shapes, losses, gradients, and a small neural network.
3. **Ghia cavity CFD**: a finite-difference streamfunction–vorticity solver for the lid-driven cavity at $\mathrm{Re} = 100$.

This lecture note supports those notebooks. It introduces the minimum fluid-mechanics vocabulary required to understand the code. The aim is not to cover a full fluid mechanics textbook in one week. The aim is to give students enough physical and numerical understanding to read a CFD code intelligently.

2 What is a fluid?

A fluid is a material that continuously deforms under applied shear stress. A solid can resist a shear stress by deforming to a new static shape. A fluid cannot maintain a static shear deformation; it keeps deforming or flowing.

Examples of fluids include air, water, oil, and gas mixtures. Liquids and gases are both fluids, even though their densities and compressibilities can be very different.

Important variables:

Symbol	Name	Meaning
ρ	density	mass per unit volume
p	pressure	normal compressive stress per unit area
$\boldsymbol{u} = (u, v)$	velocity	fluid velocity components in 2D
μ	dynamic viscosity	resistance to shear deformation
$\nu = \mu/\rho$	kinematic viscosity	momentum diffusivity
T	temperature	thermal state variable

For this Week-1 cavity problem we use a simple incompressible, constant-property fluid. That means ρ and ν are treated as constants.

3 Continuum viewpoint

Real fluids are made of molecules. CFD does not track every molecule. Instead, it treats the fluid as a continuum. A continuum field has values at every point in space, such as $u(x, y)$, $v(x, y)$, and $p(x, y)$.

This assumption is valid when the length scale of interest is much larger than molecular scales. In the Week-1 cavity, the square domain is treated as a continuous region, then sampled on a finite grid.

For AI students, the key translation is:

Fluid mechanics language	Computational language
Field $u(x, y)$	2D array of values
Grid point	pixel-like sample location
Derivative	finite-difference operation
Boundary condition	array values imposed at edges
Residual	numerical error or physics violation

4 Eulerian description

In CFD, we usually use an Eulerian description: fields are described as functions of position and time. For example,

$$u = u(x, y, t), \qquad v = v(x, y, t), \qquad p = p(x, y, t). \tag{1}$$

Here u is the horizontal velocity, v is the vertical velocity, p is pressure, x and y are spatial coordinates, and t is time.

The alternative is a Lagrangian description, where we follow individual particles. Particle methods exist, but the Week-1 CFD code uses the Eulerian grid viewpoint.

5 Velocity, acceleration, and the material derivative

Fluid acceleration is more subtle than acceleration of a single particle. At a fixed point, velocity can change with time. Also, a fluid particle can move into a region where the velocity field has a different value.

The material derivative combines these effects:

$$\frac{D\boldsymbol{u}}{Dt} = \frac{\partial \boldsymbol{u}}{\partial t} + (\boldsymbol{u} \cdot \nabla)\boldsymbol{u}. \tag{2}$$

The first term is the local time change. The second term is convective acceleration.

For a 2D velocity field $\boldsymbol{u} = (u, v)$, the convective term contains products such as

$$u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y}. \tag{3}$$

These nonlinear terms are one reason fluid mechanics is difficult and interesting.

6 Conservation of mass

The incompressible continuity equation is

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0. \tag{4}$$

This means that the velocity field is divergence-free. Fluid cannot locally accumulate or disappear.

Every incompressible CFD solver must respect Eq. (4). In pressure-based methods, pressure is used to enforce this condition. In the Week-1 streamfunction–vorticity method, the streamfunction automatically satisfies this condition.

7 Momentum equation: Navier–Stokes

For an incompressible Newtonian fluid with constant density and viscosity, the Navier–Stokes equation can be written as

$$\frac{\partial \boldsymbol{u}}{\partial t} + (\boldsymbol{u} \cdot \nabla)\boldsymbol{u} = -\frac{1}{\rho}\nabla p + \nu \nabla^2 \boldsymbol{u}. \tag{5}$$

Each term has a physical meaning:

Term	Meaning
$\partial \boldsymbol{u}/\partial t$	local unsteady acceleration
$(\boldsymbol{u} \cdot \nabla)\boldsymbol{u}$	convective acceleration
$-\nabla p/\rho$	pressure force per unit mass
$\nu \nabla^2 \boldsymbol{u}$	viscous diffusion of momentum

For Week 1 we do not directly solve Eq. (5) in pressure–velocity form. Instead, we use an equivalent 2D streamfunction–vorticity formulation that avoids pressure in the main solver.

8 Reynolds number

The Reynolds number is

$$\mathrm{Re} = \frac{UL}{\nu} = \frac{\rho UL}{\mu}. \tag{6}$$

Here U is characteristic velocity, L is characteristic length, ν is kinematic viscosity, ρ is density, and μ is dynamic viscosity.

The Reynolds number compares inertia and viscosity:

- small Re: viscous diffusion dominates; flow is smooth and strongly damped;
- moderate Re: steady laminar vortices and recirculation may appear;
- large Re: thin layers, instability, transition, and turbulence may appear depending on geometry.

In the Week-1 cavity notebook, $U = 1$, $L = 1$, and $\mathrm{Re} = 100$. Therefore,

$$\nu = \frac{UL}{\mathrm{Re}} = 0.01. \tag{7}$$

9 Boundary conditions

A PDE needs boundary conditions. For viscous flow along a solid wall, the common condition is **no slip**: the fluid velocity at the wall equals the wall velocity.

For the lid-driven cavity:

$$\text{top wall:} \quad u = 1, \quad v = 0, \tag{8}$$
$$\text{bottom wall:} \quad u = 0, \quad v = 0, \tag{9}$$
$$\text{left wall:} \quad u = 0, \quad v = 0, \tag{10}$$
$$\text{right wall:} \quad u = 0, \quad v = 0. \tag{11}$$

The moving lid drags the fluid and creates a primary vortex. The other walls stop the fluid at the boundaries.

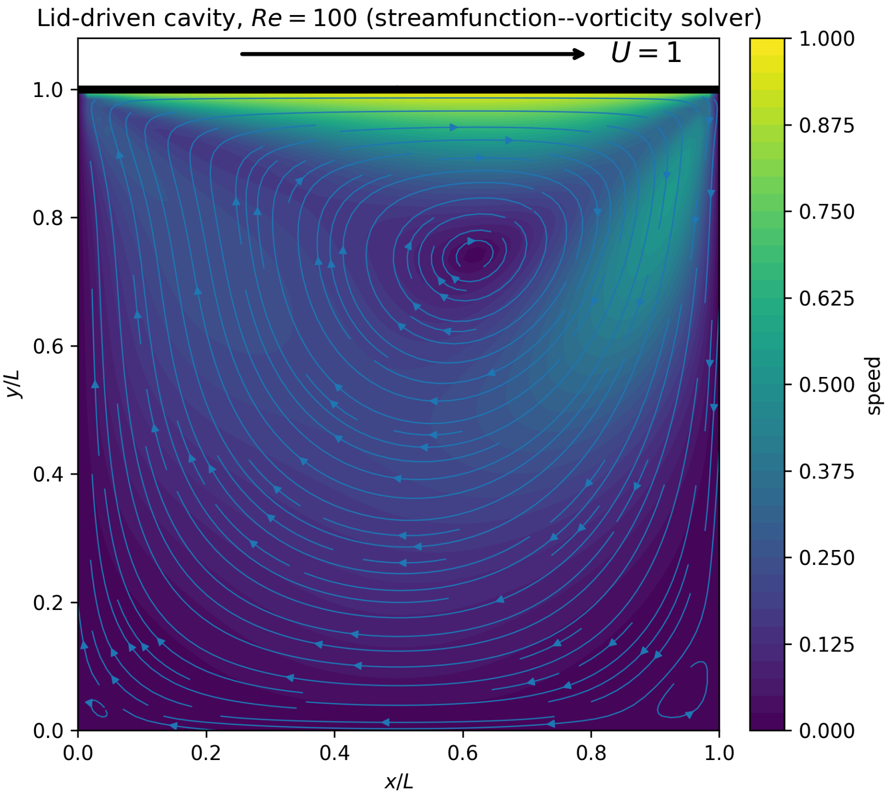


Figure 1: Lid-driven cavity streamlines computed by the finite-difference streamfunction–vorticity solver at Re = 100. The primary vortex is clearly visible; very weak lower-corner eddies may appear depending on grid resolution and convergence.

10 Streamfunction–vorticity formulation

For 2D incompressible flow, define the streamfunction ψ such that

$$u = \frac{\partial \psi}{\partial y}, \qquad v = -\frac{\partial \psi}{\partial x}. \tag{12}$$

Substituting into continuity gives

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = \frac{\partial^2 \psi}{\partial x \partial y} - \frac{\partial^2 \psi}{\partial y \partial x} = 0. \tag{13}$$

Thus incompressibility is automatically satisfied.

The vorticity is

$$\omega = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}. \tag{14}$$

For the above definition of ψ, streamfunction and vorticity are related by

$$\nabla^2 \psi = -\omega. \tag{15}$$

The vorticity transport equation is

$$\frac{\partial \omega}{\partial t} + u\frac{\partial \omega}{\partial x} + v\frac{\partial \omega}{\partial y} = \frac{1}{\mathrm{Re}}\nabla^2 \omega. \tag{16}$$

This is the method used in the Week-1 cavity notebook.

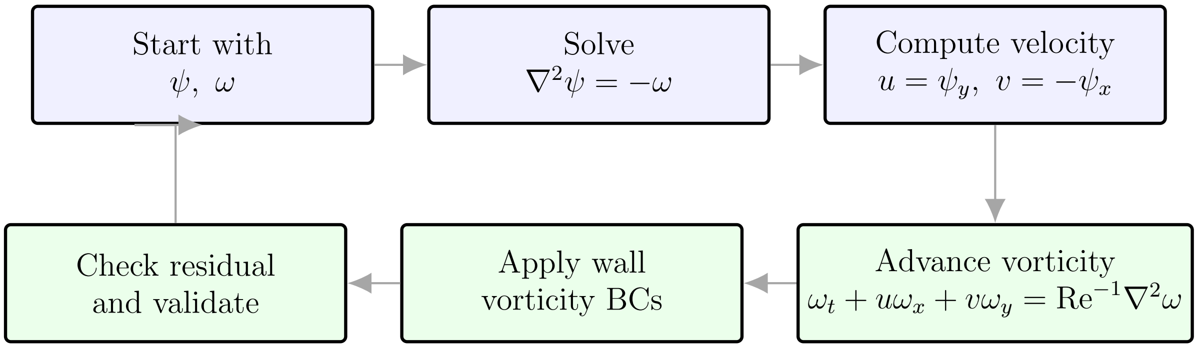


Figure 2: Educational CFD algorithm used in the cavity notebook. The solver alternates between the streamfunction equation and the vorticity transport update until the residual decreases to an acceptable value.

11 Finite differences

A computer cannot evaluate derivatives exactly at every point in a continuous domain. We approximate derivatives on a grid. For example,

$$\left.\frac{\partial f}{\partial x}\right|_{i,j} \approx \frac{f_{j,i+1} - f_{j,i-1}}{2\Delta x}. \tag{17}$$

The five-point Laplacian stencil is

$$\nabla^2 f_{j,i} \approx \frac{f_{j,i+1} + f_{j,i-1} + f_{j+1,i} + f_{j-1,i} - 4f_{j,i}}{\Delta x^2} \tag{18}$$

when $\Delta x = \Delta y$.

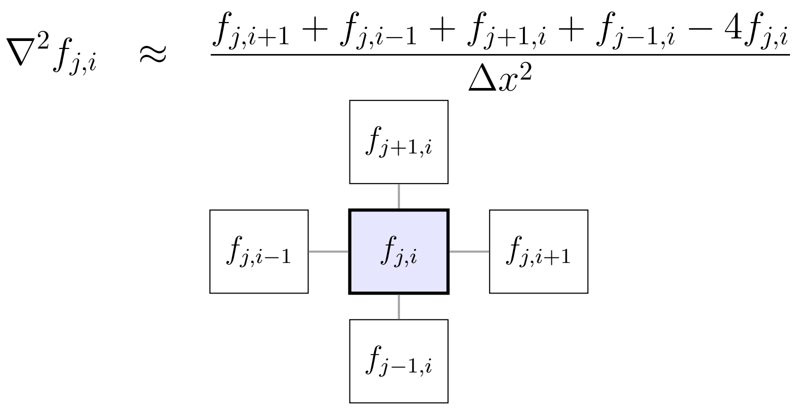


Figure 3: Five-point stencil used for finite-difference Laplacian calculations. The center point interacts with its four nearest neighbors.

12 Residuals and convergence

A residual is a numerical measure of how much the solution is still changing or how much an equation is violated. In the cavity notebook, a simple residual is based on the change in vorticity:

$$R = \frac{\|\omega^{n+1} - \omega^n\|_2}{\|\omega^{n+1}\|_2}. \tag{19}$$

A decreasing residual indicates that the flow is approaching a steady state.

This idea connects directly to machine learning. In ML, a loss measures the mismatch between prediction and target. In CFD, a residual measures the mismatch between the current field and the governing equations or steady-state condition.

13 Benchmarks: Couette, Poiseuille, and cavity

Three simple benchmarks are useful for beginners:

1. **Couette flow**: shear-driven flow between plates. It has a linear velocity profile.
2. **Poiseuille flow**: pressure-driven channel flow. It has a parabolic velocity profile.
3. **Lid-driven cavity**: recirculating flow in a square box. It is simple to define but nontrivial to solve.

The Week-1 assignment focuses on the lid-driven cavity because it connects boundary conditions, vortices, finite differences, convergence, and validation.

14 Ghia et al. validation

Ghia, Ghia, and Shin published classical benchmark data for the lid-driven cavity. For $\mathrm{Re} = 100$, they tabulated velocity values along centerlines:

- u/U along the vertical centerline $x/L = 0.5$,
- v/U along the horizontal centerline $y/L = 0.5$.

The Week-1 notebook compares the student's numerical solution with these data. This is important because a plot that "looks like a vortex" is not enough. A CFD solution must be checked quantitatively against trusted data.

15 Optional pressure recovery

The streamfunction–vorticity solver does not need pressure to compute velocity. However, pressure can be recovered after the velocity is known. A common teaching approximation uses

$$\nabla^2 p = -\left[\left(\frac{\partial u}{\partial x}\right)^2 + 2\frac{\partial u}{\partial y}\frac{\partial v}{\partial x} + \left(\frac{\partial v}{\partial y}\right)^2\right]. \tag{20}$$

Pressure is defined only up to a constant, so we subtract the mean pressure to set a gauge:

$$p \leftarrow p - \bar{p}. \tag{21}$$

This optional task is included in the cavity notebook as an extension.

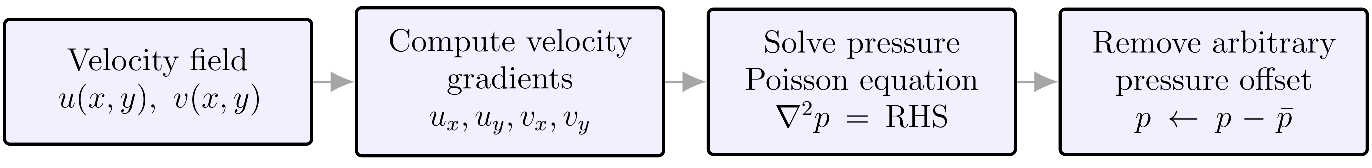


Figure 4: Optional pressure recovery from the computed velocity field. The main cavity solver computes velocity first; pressure is reconstructed afterward as an additional teaching exercise.

16 What students should know after Week 1

After completing the three notebooks, a student should be able to:

- define the main variables in incompressible flow;
- explain the role of Reynolds number;
- identify no-slip boundary conditions in code;
- interpret a 2D NumPy array as a CFD field;
- approximate a derivative with finite differences;
- understand why residuals matter;
- compare centerline velocities with benchmark data;
- explain how CFD fields become tensors for ML.

References

1. U. Ghia, K. N. Ghia, and C. T. Shin, "High-Re solutions for incompressible flow using the Navier–Stokes equations and a multigrid method," *Journal of Computational Physics*, 48, 387–411, 1982.
2. F. M. White and H. Xue, *Fluid Mechanics*, 9th ed., McGraw Hill, 2022.
3. J. D. Anderson, *Computational Fluid Dynamics: The Basics with Applications*, McGraw Hill, 1995.